

Morpheme structure phonotactics

Approaching morpho-phonological wug-tests with segment exceptionality and simpler phonotactics

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SOCIETY FOR COMPUTATION IN LINGUISTICS 2026, SAN DIEGO

An intuition: *glorp* is a better possible word of English than *lgorp*.

- A LOT of work has provided a quantitative basis for these intuitions
- The main conclusion is that those intuitions show that the speakers' knowledge makes available a metric of *similarity*

The current debate is how to model similarity:¹

- a MaxEnt grammar?
- merely segmental *n*-grams?
- *n*-grams over segments and features?

¹Hayes & Wilson (2008); Gorman (2013); Wilson & Gallagher (2018); Mayer & Nelson (2020); Kostyszyn & Heinz (2021); Dai, Mayer & Futrell (2023); Mayer (2025); Durvasula (2026)

Such studies are also used to probe productivity of lexically restricted phonological alternations: *plife* [f] ~ *plives* [v] » *pliff* [f] ~ *plivs* [v]²

- Again, quantitative basis has been established.³
- Similarity is also required (inheriting issues re: phonotactics)
- Another question pops out: similarity to *what, exactly?*

A popular implicit answer: to other words undergoing the alternation

²Becker, Nevins & Levine (2012)

³Albright & Hayes (2003), Hayes et al. (2009), Becker, Ketrez & Nevins (2011), Becker, Nevins & Levine (2012), Gouskova & Becker (2013), Moore-Cantwell & Staubs (2013), Becker & Gouskova (2016), Moore-Cantwell & Pater (2016)

The implicit answer is not innocent!

Its presupposition: it is words / morphemes that are marked as exceptional re: restricted phonological alternations.⁴

- An alternative: segments themselves are the locus of exceptionality.⁵
- Certain patterns seem to require a segment-based analysis.⁶
 - A single morpheme may contain an undergoer and a non-undergoer
 - Hebrew post-vocalic spirantization: [kavar] ~ [likbor] ‘to bury’ /kbr/

The exceptional segments lit. rarely takes nonce word studies into account.⁷

⁴Chomsky, Halle (1968) *et seq.*: see Gouskova (2012) for discussion

⁵See Zimmermann (2026) for discussion

⁶Martínez (2008), Rubach (2013), Round (2017)

⁷Usually, *a priori* criticisms are offered. See Scheer (2019) as an example.

Morpho-phonological nonce word studies are thus relevant for two debates:

- Gradient / categorical divide in phonotactics
- Segment / morpheme-based marking of exceptionality

Most existing work takes the gradient morpheme-based stance

“speakers have detailed knowledge of the phonological properties of alternating morphemes, contrary to what one might expect given theories that stipulate the identities of the alternating segments.”
(adapted from Gouskova & Becker 2013:737)

My goal today: provide a categorical segment-based alternative. Two reasons:

- To provide additional evidence in favor of categorical phonotactics
- To show that segment-based views have a way to account for those studies

The focus: Russian vowel~zero alternations (yers)

(1) a. [móx] moss.NOM.SG ~ [mx-á] moss-GEN.SG *[móx-a]

b. [nós] nose.NOM.SG ~ [nós-ə] nose-GEN.SG *[ns-á]

Question 1: what would a formal analysis look like?

- The vowels themselves differ.⁸
- The morphemes are marked as undergoers.⁹

Question 2: what do the nonce word studies show?

⁸Featurally: Lightner (1965), Halle & Matushansky (2006), Gorman (2025). Or prosodically: Kenstowicz & Rubach (1987), Scheer (2006), Scheer (2019).

⁹Gouskova (2012); Gouskova & Becker (2013); Becker & Gouskova (2016)

Gouskova&Becker (2013), Becker&Gouskova (2016): the segment-based account misses crucial generalizations regarding words that contain yers.

- Complex coda: CVCC words are less likely to alternate than (C)CVC
 - *postor*~*postr-a* » *posotr* ~ *postr-a*
- Sonority: TRVT are less likely to alternate than TTVR
 - *postor*~*postr-a* » *posrot* ~ *posrt-a*
- Syll. count: 1 syll words are less likely to alternate than 2+ syll words
 - *potor*~*potr-a* » *tor* ~ *tr-a*

These generalizations predict wellformedness judgements of nonce word alternations—how can a segment exceptionality analysis even approach this?

The idea: morpho-phonological similarity reduces to phonotactic similarity over surface *and* underlying representations.

How likely you are to accept *potor*~*potr-a*?

∝ How likely you are to accept *potr-a* as an SR and *poto_{Fr}* as a UR.

- “Speakers are more likely to choose that phoneme as the underlying representation when there are more phonologically/phonetically similar morphemes in the lexicon sharing that phoneme”
(Ernestus & Baayen 2003:29)
- The novel part is the combination with an exceptional segment theory.

Here is what you need:

- A corpus of surface representations
- A corpus of underlying representations
- A phonotactic model that evaluates a form relative to a corpus

What phonotactic model can we take?

- My goal is to show that modelling decisions of Becker and Gouskova (MaxEnt-phonotactics, morpheme exceptionality) are not the only ones compatible with their findings
- I take a categorical phonotactic model of Durvasula (2026)

Durvasula (2026) proposes a phonotactic model that is sensitive to

- Segment identity
- Feature identity (i.e., natural classes)

It is a counting model: the score of a nonce word w is # of segmental n -grams absent from a corpus plus # of valued feature n -grams absent from the corpus.

- Possible 3-gram: $\langle a, b, a \rangle$
- Possible 3-gram: $\langle +syll, -syll, +syll \rangle$
- Impossible 3-gram: $\langle \begin{matrix} -son & +son & +son \\ -syll & +syll & -syll \end{matrix} \rangle$

I compare Becker and Gouskova's (2016) sublexical model (MaxEnt + morpheme exceptionality) with the proposed UR&SR phonotactics model

- Corpus is approximated by a list of 2nd declension masculine nouns
- Feature decomposition is taken directly from B&G
- A binary $\pm float$ feature is added to generate URs from nouns in the corpus marked as containing the alternating vowel

The experimental data is taken from B&G too

- Forced choice: given a nonce item, do you accept its inflected variant?
- Each participant evaluates both with and without vowel deletion.
- Nonce items generated on the fly: not many per-item judgements.
- Nonce items differ in syllable structure and sonority patterns

As is common in the phonotactics lit. , I first present correlation metrics.

- Recall: not many per-nonce item judgements
- The data is thus aggregated by syll. type (following B&G)

Baseline is the wellformedness score of the base nonce item provided by participants. 95% CIs are bootstrapped via resampling with replacement.

Model	Pearson r [95% CI]	Spearman ρ [95% CI]	Kendall τ_b [95% CI]
B&G sublexical model (morpheme-based)	.893 [.837, .939]	.854 [.746, .915]	.673 [.566, .769]
UR & SR phonotactics (segment-based)	.844 [.778, .925]	.858 [.729, .928]	.683 [.556, .796]
Baseline	.738 [.640, .825]	.655 [.480, .758]	.432 [.293, .560]

There is a way to evaluate the models without aggregation (AUC)

- All observations are of the form (item, 0) or (item, 1).
- A good model should score x as less likely than y if there is a pair of observations $(x, 0)$ and $(y, 1)$.
- Our metric is a proportion of such pairs the model gets right.

The results suggest that the two models are at least on equal footing.

Model	AUC (item-weighted) [95% CI]
UR & SR phonotactics (segment-based)	.652 [.629, .673]
B&G's sublexical model (morpheme-based)	.637 [.615, .659]
Baseline	.613 [.592, .634]

Another way to compare the models is to bootstrap the difference between the AUC scores of the models.

The results suggest that the current proposal outperforms the sublexical model

Comparison	Mean Δ AUC [95% CI]
UR & SR phonotactics vs. Sublexical	.014 [.002, .026]
UR & SR phonotactics vs. Baseline	.039 [.021, .056]

Consider a trigram perspective, combines with the observation that the alternating vowels are last vowels in a word/morpheme (Gouskova 2012).¹⁰

- Take the complex coda effect: CVCC# nonce items alternate poorly
- Such items would have V_F CC UR trigrams. Only complex coda items would: since V_F is necessarily the last vowel.
- The absence of V_F CC UR trigrams explains the complex coda effect.
- See the paper for the discussion of the other two generalizations.

The observations are explained by the surroundings that V_F may have.

¹⁰It is a consequence of alternation-driven learning (Scheer 2019): only the last vowel alternates.

Interpretation of morpho-phonological wug-tests needs explicit footing on the notion of similarity to existing lexicon.

- How is similarity quantified? Q. of a phonotactic model
- Similarity to what, exactly? Q. of a theory of exceptionality

This talk revisited the results argued to provide evidence for model combining a morpheme-based theory exceptionality with MaxEnt-based phonotactics (the sublexical model of Becker & Gouskova).

- A categorical phonotactic model (Durvasula 2026)
- Phonotactic similarity to URs and SRs (cf. Ernestus, Baayen 2003)

Takeaway: wug-tests do not adjudicate between different proposals re: Russian vowel-zero alternations.

The model should be tested on further data

- A proper list of Russian morphemes, not limited by 2nd decl. masc.
 - The complex coda effect has two exceptions (Scheer 2019):
mest ~ *mst-it* ‘revenge’ and *lest* ~ *lst-it* ‘flatter’.
- Other morpho-phonological wug-test results

Other modelling decisions should be explored

- Alternative categorical phonotactic models
 - Feature-insensitive models are insufficient (Appendix A)
 - BUFA (Swanwon, Rawski, and Heinz 2026): to be tested
- Varying feature decompositions (see the next slide)

Durvasula's (2026) model relies on segments and features

- Segment-insensitivity does not lead to a major performance drop (see Appendix B of the paper)
- What matters is whether there is a natural class that only includes the alternating vowels
 - Many proposals treat them as unmarked relative to the non-alternating vowels — suggesting their insufficiency, at least given a particular linking hypothesis to nonce word tasks.
- Gorman, Reiss (2025) provide an *a priori* reason to suspect that this only holds of exceptional deletion processes — further research is needed to determine whether it is the case.

-  Albright, Adam & Bruce Hayes. 2003. Rules vs. analogy in english past tenses: a computational/experimental study. *Cognition* 90(2). 119–161.
-  Becker, Michael & Maria Gouskova. 2016. Source-oriented generalizations as grammar inference in russian vowel deletion. *Linguistic inquiry* 47(3). 391–425.
-  Becker, Michael, Nihan Ketrez & Andrew Nevins. 2011. The surfeit of the stimulus: analytic biases filter lexical statistics in turkish laryngeal alternations. *Language* 87(1). 84–125.
-  Becker, Michael, Andrew Nevins & Jonathan Levine. 2012. Asymmetries in generalizing alternations to and from initial syllables. *Language* 88(2). 231–268.
-  Dai, Huteng, Connor Mayer & Richard Futrell. 2023. Rethinking representations: a log-bilinear model of phonotactics. *Proceedings of the Society for Computation in Linguistics* 6.

-  Durvasula, Karthik. 2026. A closer look at what/how we can learn from computational modelling of phonotactics. *Ms. University of Michigan*.
-  Ernestus, Mirjam & R Harald Baayen. 2003. Predicting the unpredictable: Interpreting neutralized segments in Dutch. *Language* 79(1). 5–38.
-  Gorman, Kyle. 2013. *Generative phonotactics*. University of Pennsylvania dissertation.
-  Gorman, Kyle. 2025. *A Logical Phonology of some "minor rules" of Polish*.
-  Gouskova, Maria. 2012. Unexceptional segments. *Natural Language & Linguistic Theory* 30(1). 79–133.
-  Gouskova, Maria & Michael Becker. 2013. Nonce words show that Russian yer alternations are governed by the grammar. *Natural Language & Linguistic Theory* 31(3). 735–765.
-  Halle, Morris & Ora Matushansky. 2006. The morphophonology of russian adjectival inflection. *Linguistic inquiry* 37(3). 351–404.

-  Hayes, Bruce & Colin Wilson. 2008. A maximum entropy model of phonotactics and phonotactic learning. *Linguistic inquiry* 39(3). 379–440.
-  Hayes, Bruce et al. 2009. Natural and unnatural constraints in hungarian vowel harmony. *Language* 85(4). 822–863.
-  Kenstowicz, Michael & Jerzy Rubach. 1987. The phonology of syllabic nuclei in slovak. *Language*. 463–497.
-  Kostyszyn, Kalina & Jeffrey Heinz. 2021. Categorical account of gradient acceptability of word-initial polish onsets. In *Proceedings of the annual meetings on phonology*.
-  Lightner, Theodore McGraw. 1965. *Segmental phonology of modern standard russian*. Massachusetts Institute of Technology dissertation.
-  Martínez, Michal Temkin. 2008. Exceptionality and variation in modern hebrew spirantization. In *Wccfl* 27, 413–421.
-  Mayer, Connor. 2025. Reconciling categorical and gradient models of phonotactics. *Society for Computation in Linguistics* 8(1).

-  Mayer, Connor & Max Nelson. 2020. Phonotactic learning with neural language models. *Society for Computation in Linguistics* 3(1).
-  Moore-Cantwell, Claire & Joe Pater. 2016. Gradient exceptionality in maximum entropy grammar with lexically specific constraints. *Catalan Journal of Linguistics* 15. 53–66.
-  Moore-Cantwell, Claire & Robert Staubs. 2013. Modeling morphological subgeneralizations. In *Proceedings of the annual meetings on phonology*.
-  Round, Erich. 2017. Phonological exceptionality is localized to phonological elements: the argument from learnability and yidiny word-final deletion. *On looking into words (and beyond): Structures, relations, analyses*. 59–97.
-  Rubach, Jerzy. 2013. Exceptional segments in polish. *Natural Language & Linguistic Theory* 31(4). 1139–1162.
-  Scheer, Tobias. 2006. How yers made lightner, gussmann, rubach, spencer and others invent cvcv. *Studies in constraint-based phonology*. 133–207.

-  Scheer, Tobias. 2019. On the difference between the lexicon and computation (regarding slavic yers). *Linguistic inquiry* 50(1). 197–218.
-  Wilson, Colin & Gillian Gallagher. 2018. Accidental gaps and surface-based phonotactic learning: a case study of south bolivian quechua. *Linguistic Inquiry* 49(3). 610–623.
-  Zimmermann, Eva. 2026. Morphemes in phonology. In Paul de Lacy & Adam Jardine (eds.), *The cambridge handbook of phonology* (2nd ed.). Cambridge University Press.